Anonymization Techniques for Protecting Privacy: A Survey

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***Abstract*— Anonymization is one of fruitful privacy protection technique used in various technology fields such as data mining, cloud computing, big data to secure very sensitive data against third party. In today's world, the value and the amount of data is increasing, hence the protection of data against all possible threats are equally necessary. This paper focuses a brief on data anonymization and differential privacy techniques. Various anonymization techniques which are researched by various researchers across various fields have limitations such as communication and computation cost overhead, accuracy of results after data Anonymization and possibility of different types of attacks. The paper discussed all these issues and their counter-measures through readings of various papers. Finally, this paper presents detailed discussion about existing anonymization techniques (Data anonymization and differential privacy), their comparative analysis by leaving a footprints of future research directions.**

Keywords*— *Anonymization, Privacy, Data Anonymization, Differential Privacy

I. Introduction

Security and Privacy concerns of any type of data is major issue in today's technology-driven world. The term privacy is defined as an action where the data is kept hidden from either anonymous user, server to avoid use of malpractice of the data. In the world of distributed environment, the processing and storage of such data is also done so dynamically at dynamic location keeping the location transparency in mind. In such scenario, privacy of data is important concern. Primarily there are some privacy techniques that are used to hide data viz. Anonymization, Role Based Access Control, Attribute Based Encryption etc. Data has different phases during its lifecycle: storage, transition, transfer and processing. Existing privacy preserving techniques are still in maturing stages and strong privacy protection is still an open research problem.

Anonymization is one of the most fruitful and promising techniques that have been implemented over a period of the year. Through various research and products, anonymization techniques are fruitful against following attacks:

- Homogeneity Attack: in this type of attack if all the records of a sensitive data is anonymized, then it is easy for attacker to get access on sensitive information.to overcome this attack we have L-diversity principle used.
- Similarity Attack: In this case , if all sensitive information of all the records may not be exactly same but may be similar then ,attacker may gain sufficient knowledge.
- Background knowledge Attack: If attacker has some knowledge about sensitive value he/she can use this background knowledge to understand actual sensitive information. E.g. In case of EHS if attacker knows that the target has some health issue related heart. Attacker can use this knowledge to understand exact health information of a target.
- Probabilistic Inference Attack: In particular sensitive values are occurring frequently then such anonymized data become prone to probabilistic inference attack. E.g. In case of EHS if heart issues are frequently occurring in anonymized group then the probability of heart issue increases for target patient.

Anonymity methods are designed to make data, user, and communication anonymous and un-linkable. As discussed in [1] following are different types of Anonymization:

Data Anonymization: Data anonymity breaks the relation between user and user's confidential data.

User Anonymization: User anonymity provides privacy about user's identity (e.g. name or SSN).

Communication Anonymization: Communication anonymity hides the relation between sender and receiver.

Unlinkability: Unlinkability anonymizes the relation between sender and data

The paper contribution in this paper is as follows. Section II has detailed related work where different types of anonymization and their methods are discussed. Section III discusses about data anonymization techniques along with the current work in the same field. Section IV discusses differential privacy along with the current work done in the same field. Finally section V concludes this survey.

II. Related Work

The different type of Anonymization and their methods are categorized in Table 1.

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TABLE I. ANONYMIZATION METHODS FOR PRIVACY PROTECTION

Anonymization Type → / Anonymization Method ↓	Data	User	Communication	Unlinkability
K-Anonymity[2]	*Yes*	*Yes*	*No*	*No*
L-diversity[3]	*Yes*	*Yes*	*No*	*No*
T-closeness[4]	*Yes*	*Yes*	*No*	*No*
Pseudonymization: Tokenization/ Hashing[5-10]	*Yes*	*Yes*	*No*	*Yes*
Differential Privacy [11,12]	*Yes*	*Yes*	*No*	*No*
Onion Routing[13-18]	*No*	*No*	*Yes*	*No*

K-anonymity [2]: It uses data aggregation, suppression, generalization methods in order to generate k-anonymous records of the same data. Thus k-Anonymization method generates data set in which each person in the database cannot be distinguished from at least k-1 individuals.

L-diversity [3]: It is considered as an advanced and extended version of k-anonymity. From the available literature survey it is known that: K-anonymity is prone to interference attack in which the attacker to relate the visible value to the real value. L-diversity protects anonymity by giving every attribute at least L different values.

T-closeness [4]: T-closeness is another extension of k-anonymity which also solves the problem of attribute disclosure. This is powerful technique as compare to l-diversity.

Pseudonymization [5-10]: Instead of removing all the identifying information like anonymization, pseudo-randomization focuses on decreasing the linkability of the given dataset with original identity information.

Following are examples of Pseudonymization techniques:

- *Hash functions*. One of the major characteristic of the hash functions is that they are fast computable. In hash functions there is mapping of data of dataset to relevant codes of a fixed size takes place.
- *Tokenization*. In tokenization there is a substitution of data components with a non-sensitive equivalent. The equivalent is called as token. The token serves as an unique identifier and it is a pointer that traces back to the original data.

Differential privacy [11, 12]*:* Differential privacy achieves privacy by adding noise to the data. Differential privacy is applied to data before sharing the data third party e.g. sharing medical data with researchers for doing clinical research. This provides patients privacy by hiding their identity but allows learning statistical results from the data.

Onion Routing [13-18]: Onion Routing provides communication privacy over a public network. It basically provides anonymous connections which are strongly resistant to both eavesdropping and traffic analysis.

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III. DATA ANONYMIZATION

Today popularity and wide adoption of distributed technologies such as cloud computing and IoT have caused major distribution of data at different geographical locations. Also, data has become major asset for business organization and thus data is used by data scientists for performing different data analytics and extracting knowledge from big data.

A. Introduction

Data Anonymization is one of the promising privacy preserving techniques. As shown in Fig. 1 data Anonymization process is applied to data before handing over data to third party such as medical researcher so as to make data anonymous and protect the data privacy.

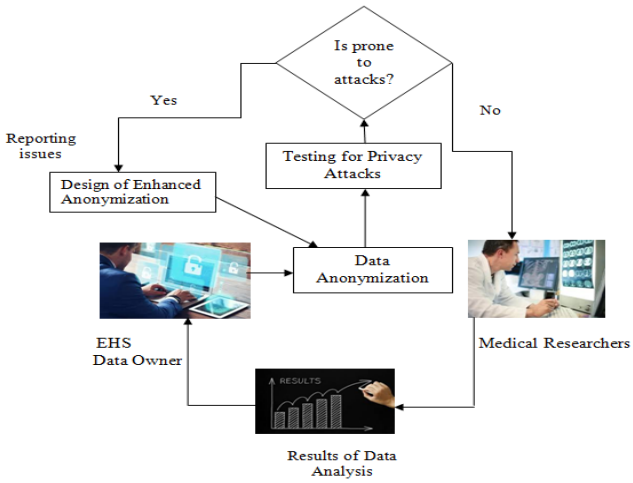

Fig. 1. Data Anonymization Process: Basic Architecture

Fig. 1 Presents basic architecture of data Anonymization process which includes two entities data owner and third party data processing service provider. In this Data owner can be Electronic Healthcare service provider who owns patients health data and third party data processing service provider can be medical researchers. It is necessary to make sensitive patient health information anonymous before handing over it to semi trusted data processing service provider.

B. Techniques

K-anonymity [2]: Generalization and suppression are common methods used for generating k-similar records as shown in Table 2. These k-records become indistinguishable after K-Anonymity application. Thus third party cannot exactly identify individual record from these k-similar looking records. Thus k-anonymity successful achieves prevention of identity disclosure and linkage attack. Identity disclosure means identifying an individual's record and linkage means linking individual to particular sensitive information. Generalization is NP-hard problem and suppression leads to loss of information. K-anonymity is frequently used for healthcare data Anonymization.

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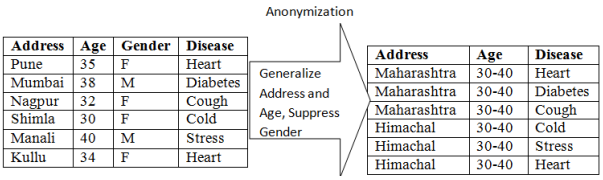

Address	Age	Gender	Disease
Pune	35	F	Heart
Mumbai	38	M	Diabetes
Nagpur	32	F	Cough
Shimla	30	F	Cold
Manali	40	M	Stress
Kullu	34	F	Heart

Address	Age	Disease
Maharashtra	30-40	Heart
Maharashtra	30-40	Diabetes
Maharashtra	30-40	Cough
Himachal	30-40	Cold
Himachal	30-40	Stress
Himachal	30-40	Heart

Fig. 2. 3-Anonymous Data from original data.

Clustering based K-anonymity [19, 20 and 21]: Clustering based k-anonymity methods proposed by researchers to speed up Anonymization and to minimize the information loss in Anonymization process.

K-Anonymization is prone to attacks such as homogeneity attack, similarity attack, background attack and probabilistic inference attack.

L-diversity [2]: L-diverse technique was introduced to prevent attacks on k- anonymous data. It is necessary to have diversity in values of sensitive attributes grouped in k-anonymity. If all k records in a group posses same sensitive information then one can easily attack on such data and gain information about an individual and thus leads to violation of individual's privacy. If Anonymization maintains l-diversity then anonymized data remains resistant to different attacks.

As shown in figure 3 , after anonymization if k records in any group contain same sensitive information then k-anonymous data remain prone to different attacks. In second part of figure 3, records contains different sensitive information for all 3 records thus makes it 3-diverse. L-diversity is prone to probabilistic inference attacks.

Address	Age	Gender	Disease
Maharashtra	30-40	Maharashtra	Heart
Maharashtra	30-40	Maharashtra	Heart
Maharashtra	30-40	Maharashtra	Heart
Himachal	30-40	Himachal	Cold
Himachal	30-40	Himachal	Cold
Himachal	30-40	Himachal	Cold

Address	Age	Disease
Maharashtra	30-40	Heart
Maharashtra	30-40	Diabetes
Maharashtra	30-40	Cough
Himachal	30-40	Cold
Himachal	30-40	Stress
Himachal	30-40	Heart

Fig. 3. L-Diversity for the sensetive data

T-closeness: T-Closeness overcomes drawback of l-diversity and guarantees distribution of sensitive values in equivalence classes same as that of original data. T-closeness is practically difficult to achieve.

S-diversity: S-diversity overcomes similarity attacks in worst case scenario by using clustering algorithm.

C. Comparison

TABLE II. COMPARISON OF DATA ANONYMIZATION TECHNIQUES

Paper	Technique	Advantages /Attacks prevented	Limitations / Prone to attacks
[31]	two-phase top-down specialization (TDS) approach	Authors have proposed new technique of data anonymization on cloud using map reduce model. Larger number of cloud service requires the diverse and big data which is sharable and brings privacy concerns. Traditional data anonymization methods does not fulfill the requirement of this aspect of data privacy. After the implementation of proposed approach, the scalability and efficiency is significantly improved.	• It is a fact that, increasing volume of a data on a cloud is the big challenge for privacy preservation , data analysis of a data , hence only top down approach may not be always appropriate approach to investigate. • Authors have not mentioned the bottom up approach to investigate the privacy data on cloud, which is left for future scope. • Though the architecture is quite scalable from the results, the scalable awareness analysis is missing.
[32]	Blockchain and off blockchain storage	Authors have constructed personal data management suing blockchain and off blockchain storage to ensure strong privacy. Authors have also discussed the future enhancement of blockchain implementation for trusted computing. Author has claimed their research is susceptible to Sybil attacks.	• Blockchain has very much legal issues and its acceptability rate in a world is significantly less . • Author has suggested that laws and regulations must be programmed in the blockchain itself so that trust level can be increased. • There is no procedure / methodology explained for susceptibility of Sybil attacks.
[33]	l-site-diversity	In this paper, author has developed decentralized protocol for horizontal data partitioned database to enable privacy across distributed environment. A new approach called as l–site-diversity is used for developing protocol. The work is susceptible for homogeneity attack on distributed databases.	• The author has addressed only one/two privacy principles and anonymization protocols, in distributed environment, as the data is dispersed; we need more anonymization protocols to address the issue of privacy.
[34]	k-Diversity and l-diversity and space mapping technique	The existing technique of data anonymization have a problem of information loss and inaccuracy of data due to generalization, computation cost. Author have proposed k-diversity and l-diversity with space mapping technique to achieve faster execution time and less information loss.	• In future the author has promised that the dual problem can be studied i.e. form maximum value of information loss , what is the correct degree of privacy (either k or l) • No attack is mentioned in the paper.

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[35]	Trivial sanitization	Since k-diversity does not guarantees privacy whereas l-diversity requires distribution of attributes in dataset with high entropy – the paper uses trivial sanitization which has better utility than k and l-diversity. The research is susceptible for unsorted matching attack [38]	• Since the chosen dataset is not real-world dataset , research requires more fruitful strategies of trivial sanitization on real world data-set. • The algorithm which is designed by author (micro data sanitization algorithm) is not applicable to both i.e. utility and privacy.
[36]		In this paper author has reviewed the problem of protecting privacy on set-valued data . The author have developed a strategy k^m –anonymity which focuses on generalization of data instead of suppression. The proposed greedy heuristics algorithms are experimentally evaluated using real database. Author has claimed their research is susceptible to background attacks.	• The model developed by author is only applicable to k-diversity and not to l-diversity. • Information loss is important factor in privacy which needs to be addressed and hence recoding technique needs to be refine.
[37]	Survey of clustering-based approaches and graph modification approaches.	This survey paper studies the anonymization in social media data. The paper also identifies challenges in privacy preserving publishing of social network data. Author have done categorization on anonymization techniques in privacy preserving publishing of social network data. The author has decided to classify into clustering-based approaches and graph modification approaches. The paper examines all the aspect of background knowledge based attack on social media with the help of case study.	• The paper studies various anonymization techniques for publishing the data on social media. • The author has concluded that the privacy of social media data problem must be formulated based on three important factors : privacy, background knowledge and data utility. • Since the data available in social media is complex and interrelated, privacy preserving techniques must be more stronger and fruitful.

IV. Differntial Privacy

A. Introduction

Differential privacy provides generalized statistical analysis without revealing the identity of authenticated user over a data. As it hides the individual identity of a data of a user, it is consider as an extra layer of security against re-identification attacks and attacks happens on auxiliary data.

According to the book [19], to achieve differential privacy, random noise is added into query results which help to add or remove the arbitrary record of an individual user into dataset. The differential privacy is used in many applications like social-network analysis, Online advertising [20], securing health data [21] etc. According to the article [22], uber has recently released an open source project on differential privacy. The privacy team of uber has built software which has scalable integration with development workflow using simple RPC.

Over the period of year, there are lot of research work has been done on differential privacy. Many researchers are attempting machine learning and data mining techniques to achieve differential privacy. The contribution to each researcher is described in table III.

B. Techniques

The privacy cannot be applied without use of deterministic functions. Differential privacy also guarantees the sensitive data and the related computing function operating on that data is secret. Another fact associated with differential privacy is to make use of randomized function as it provides plausible deniability [23]. There are majorly three key approaches of differential privacy, they are:

Input Perturbation [24] – this approach of differential privacy adds random noise to the sensitive dataset. The main advantage of input perturbation is the reproducibility of the noisy dataset. i.e. the we can obtain the separate noisy dataset from the main dataset , Inspite of the same we can achieve the privacy. The other researchers can use the same noisy dataset to apply on another application to produce other results. [24]

Output Perturbation [25]: in this approach, the noise is added after the results of the computation of an algorithm. Laplace example can be an example of output perturbation. As the noise is added after the results of output , therefore it is completely dependent on the behavior of the algorithm.

Algorithm Perturbation [19] : in this approach , the noise is added during the computation of the results of dataset. Unlike the input perturbation , the amount of noise is not algorithm-specific but it is utility-specific.

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C. Comparison

Table III. Comparison of Differential Privacy Techniques

Paper	Technique used in the paper	Advantages /Attacks prevented	Limitations / Prone to attacks
[26]	Algorithm Perturbation	The paper suggests differential privacy for neural networks and machine learning algorithms. Machine learning/neural network algorithm often requires large dataset which may contain sensitive information. These models should not expose the confidential information. The research therefore suggests cost effective, less complex framework to train the neural network for larger sensitive dataset. The research is susceptible for model-inversion attack.	• The accuracy of the experimental results need to be improved as stated in the future direction of the paper.
[27]	Output perturbation	The paper covers bridging the gap by analyzing k-anonymization methods and it show how they fail to provide re-identification attack. After performing experiment, the research gives alternate solution to output perturbation. A random sampling is added to achieve differential privacy at the beginning and pruning results, which archives the strong layer of privacy. This research is susceptible for re-identification attack due to random sampling.	• The paper only targets the output perturbation (i.e. the random sampling is added after query results). • As the random sampling is used to amplify the privacy protections for algorithm, it can be apply for input perturbation also. • Differential privacy require δ to be very small to provide sufficient privacy protection when publishing micro data. As author have used f-smooth (ϱ, δ)-differential privacy which can be satisfied the random sampling. • The other methods can be tested to satisfy (ϱ, δ), which are not tested in this paper.
[28]	Output Perturbation	The paper examines some practical challenges of applying differential privacy on health data. The paper also proposed a model to examine the health care data using differential privacy. The research examines the re-identification attack on the sensitive information of health data of a patient.	• Authors have stated that, health care data still needs to be disclosed, also specific considerations needs to be addressed. • If done so, the more methods can be fruitful to implement differential privacy. • The paper speaks about re-identification attack but the methods to prevent this attack are not addressed.
[29]	Algorithm Perturbation	This paper addresses the problem by considering privacy and the nature of algorithm simultaneously by applying data mining algorithm - Decision tree. The improved algorithm achieves higher level of privacy and accuracy of data on fewer learning sample that are taken for the experiments.	• After the execution of experiment, the variance of data is relatively high. However author have suggested to use other stopping rules but it is not experimentally based in this paper. • The paper speaks about auxiliary attack but the methods to prevent this attack are not addressed.
[30]	Output Perturbation	The paper proves the fact that privacy of an individual using location pattern mining does not increase the risk factor. For the experiment, author have used statistical and synthetic dataset as it is feasible for location pattern mining algorithm. This research is susceptible for auxiliary attack due to random sampling.	• The paper uses synthetic data to achieve differential privacy; it would be more feasible if author can use the real time data.

V. Conclusion

After the detailed study of data anonymization and differential privacy, it can be concluded that there has been a lot of work done in the area of privacy in the applications like healthcare, IoT, Big data analytics. Despite of this there are many attacks which are still unanswered in case of privacy issues. The proposed methods of privacy in data anonymization and differential privacy has less accuracy results. The privacy issue in a new area like blockchain is still a major challenge for today's world .

The future aspects of data anonymization includes a new techniques of schemes which are universal for any kind of data and also which addresses all kind of privacy attacks possible. The future enhancement in differential privacy includes the improvisation in the parameter accuracy and efficiency of the system.

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